

SIGNIFICANCE OF PLANT CULTIVAR AS A PREDICTOR OF INSECT-CROP INTERACTIONS

Priority Area(s): Agriculture, Entomology, Pollination

Overview

Many crops dependent on pollination for proliferation are not reaching optimal yield capacity. The cultivar of the plant has been shown to influence the frequency of pollinator insect visitation, though referenced data is limited in both the plant and insect species studied — highbush blueberry and bees, respectively. A better understanding of which cultivars are preferred by various pollinator insects can provide farmers with information for strategic crop selection.

Goals and Objectives

This project aims to further establish the causal effect between plant cultivar and pollination frequency. Through regression modeling and post-fit analysis, equations for more accurate predictions on future crop yield may be developed.

Research Plan (task-based with timeline)

Task 1: Fitting different regression models → This task implements the modules of Data Preparation and Model Selection to work with multiple types of regression models in the pursuit of mapping data points to an estimated equation.

Task 2: Analyzing residuals of fitted regression models → This task implements the modules of Model Evaluation and Critical Thinking to determine through diagnostics which of the regression models best reflects the real-life relationship between plant cultivar and pollination frequency.

Evaluation Metrics

Monthly checkpoints on progress through the tasks can be determined via code submissions, graded on functionality and relevance to the project. A summary or explanation of purpose for each section of code should be included.

Anticipated Outcomes & Impacts

This project's anticipated outcome is to obtain the best-fit equation for predicting the frequency of bee pollination of highbush blueberry crops given the cultivar. A potential impact is agricultural use in optimizing blueberry crop yield.

Follow-up Funding/Follow-up Research/Growth Plan

I plan to continue this line of research into broader interactions between animal species and their environments. This project will lay the foundation for more extensive data fitting and analysis.

REFERENCES

[1] Eeraerts, Maxime (2024). *Pollination deficits and their relation with insect pollinator visitation are cultivar-dependent in an entomophilous crop*. Figshare. Dataset. <https://doi.org/10.6084/m9.figshare.25583262.v1>